

The $2p$ order Heisenberg-Pauli-Weyl uncertainty principles related to the offset linear canonical transform

Jia-Yin Peng ^{a,b}, Bing-Zhao Li ^{a,b,*}

^a School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, 100081, China

^b Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing, 100081, China

ARTICLE INFO

Keywords:

Uncertainty principle

Heisenberg-Pauli-Weyl uncertainty principle

Offset linear canonical transform

ABSTRACT

The uncertainty principle is one of the fundamental tools for time-frequency analysis in signal processing, revealing the intrinsic trade-off between time and frequency resolutions. With the continuous development of various advanced time-frequency analysis methods based on the Fourier transform, investigating uncertainty principles associated with these methods has become one of the most interesting topics. This paper studies the uncertainty principles related to the offset linear canonical transform, including the Plancherel-Parseval-Rayleigh identity, the $2p$ order Heisenberg-Pauli-Weyl uncertainty principle, the Heisenberg-Weyl uncertainty principle and the sharpened Heisenberg-Weyl uncertainty principle. Numerical simulations are also proposed to validate the derived results.

1. Introduction

Heisenberg inequality [1–3] was first proposed by the German physicist Heisenberg in 1927, which indicates that in a quantum system, the position and momentum of a particle cannot be precisely measured simultaneously. In 1928, H. Weyl and W. Pauli demonstrated that the energy of a signal and that of its Fourier transform (FT) also satisfy the Heisenberg inequality. The FT [4–6] is one of the most significant analytical tools in signal processing, enabling the transformation of signals from the time domain to the frequency domain. Consequently, the Heisenberg inequality reveals an inherent limitation in the simultaneous concentration of energy in both the time and frequency domains [7], which is commonly recognized as the Heisenberg uncertainty principle in signal processing. This principle states that the time duration $\Delta_{f(t)}$ of a signal $f(t)$ and the frequency bandwidth $\Delta_{\hat{f}(\xi)}$ of its FT $\hat{f}(\xi)$ is no less than $1/2$ [8].

This uncertainty principle conveys two valuable insights [9]. First, the resolutions in the time and frequency domains cannot be simultaneously increased without bound, as their product is constrained by a nonzero lower limit. Second, there exists a trade-off relationship between time and frequency resolution, meaning that improving frequency resolution inevitably leads to a reduction in time resolution, and vice versa. Time resolution and frequency resolution are key parameters in signal detection. High time resolution facilitates precise localization, while high frequency resolution enables accurate estimation of spectral

components, such as velocity or modulation characteristics. Therefore, the uncertainty principle has become one of the important topics in signal processing [10–12].

Building on the above research, John Michael Rassias proposed the generalized Heisenberg-Pauli-Weyl (HPW) uncertainty principle in 2004 [7]. In the following year, he presented the Heisenberg-Weyl (HW) uncertainty principle [13]. To obtain a more precise bound for practical applications, Rassias proposed the sharpened Heisenberg-Weyl (SHW) uncertainty principle in the Fourier analysis in 2006 [14]. The above uncertainty principles are all established based on the FT [4–6], which can convert time domain signals into the frequency domain to facilitate the analysis of their frequency characteristics. However, for non-stationary signals, the FT falls short in providing optimal time-frequency resolution, as it cannot effectively capture the temporal variations of the spectrum. To address this limitation, a range of advanced time-frequency analysis tools such as the fractional Fourier transform (FrFT) [15–17], the linear canonical transform (LCT) [18,19] and the offset linear canonical transform (OLCT) have been developed.

The OLCT is the generalization of the FT, FrFT and LCT [20,21]. It has six parameters (a, b, c, d, τ, η) , which makes it more flexible. In recent years, various uncertainty principles based on the FrFT [22–24], LCT [25–27] and OLCT [28–30] have been extensively investigated. However, the $2p$ order HPW uncertainty principles in the OLCT domain remain unexplored, highlighting the need for further research in this area. The OLCT has demonstrated considerable theoretical and practical po-

* Corresponding author.

E-mail addresses: 18764988153@163.com (J.-Y. Peng), li_bingzhao@bit.edu.cn (B.-Z. Li).

<https://doi.org/10.1016/j.dsp.2025.105746>

Available online 19 November 2025

1051-2004/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

tential in applications such as time-frequency analysis, filter design, and target detection [31]. Therefore, studying the uncertainty principle associated with the OLCT is of both theoretical and practical importance.

In this paper, three types of the Heisenberg uncertainty principles related to the OLCT are proposed. Section 2 is the preliminaries of this work. In Section 3, we extend the Plancherel-Parseval-Rayleigh (PPR) identity, the $2p$ order HPW uncertainty principle, HW uncertainty principle and the SHW uncertainty principle in OLCT domain. Section 4 gives experimental validations of the proposed theoretical results. Section 5 concludes this paper.

2. Preliminaries

In this section, we briefly introduce the definition of the OLCT and the fundamental concepts required for deriving the main results. Additionally, we review the $2p$ order Heisenberg uncertainty principle based on the FT.

2.1. The definition of the OLCT

The OLCT of a signal $f(t)$ with parameter $J = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{vmatrix} \tau \\ \eta \end{vmatrix}$, is defined as follows Pei and Ding [20]

$$O_J^f(\xi) = O_J[f(t)](\xi) = \begin{cases} \int_{-\infty}^{+\infty} f(t) K_J(t, \xi) dt, & b \neq 0 \\ \sqrt{d} e^{j[\frac{cd(\xi-\tau)^2}{2} + \xi\eta]} f(d\xi - d\tau), & b = 0 \end{cases}, \quad (1)$$

where $K_J(t, \xi) = \frac{1}{\sqrt{2\pi b}} e^{j[\frac{a}{2b}t^2 - \frac{1}{b}t(\xi-\tau) - \frac{1}{b}\xi(d\tau - b\eta) + \frac{d}{2b}(\xi^2 + \tau^2)]}$ and parameters $a, b, c, d, \tau, \eta \in \mathbb{R}$, $ad - bc = 1$. When the parameter $J = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{vmatrix} 0 \\ 0 \end{vmatrix}$, the OLCT becomes LCT. When the parameter $J = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{vmatrix} 0 \\ 0 \end{vmatrix}$, the OLCT becomes FrFt. When the parameter $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \end{vmatrix}$, the OLCT becomes FT.

2.2. Some identities

This section presents several fundamental equations used throughout this paper. We begin by introducing a differential identity [7].

Identity 1. Suppose that $f(t)$ is a signal with $t \in \mathbb{R}$, then for any fixed but arbitrary integer $k \in \mathbb{N}^+ = \{1, 2, \dots\}$, the following identity holds

$$f(t) \overline{f^{(k)}(t)} + f^{(k)}(t) \overline{f(t)} = \sum_{l=0}^{\lfloor \frac{k}{2} \rfloor} (-1)^l \frac{k}{k-l} \binom{k-l}{l} \frac{d^{k-2l}}{dt^{k-2l}} |f^{(l)}(t)|^2, \quad (2)$$

where $f^{(l)}(t) = \frac{d^l}{dt^l} f(t)$ is the l -th derivative of $f(t)$, $l \in \mathbb{N} = \{0, 1, 2, \dots\}$, with $0 \leq l \leq \lfloor \frac{k}{2} \rfloor$, and $\lfloor \cdot \rfloor$ denotes the floor function (i.e., the greatest integer less than or equal to its argument), $\overline{(\cdot)}$ is the conjugate of $(\cdot)$.

Next, we introduce a Lagrange type differential identity [7].

Identity 2. Suppose that $f_\alpha(t) = e^{-j\alpha t} f(t)$, with $\alpha = 2\pi\xi_m$, for any fixed but arbitrary real constant ξ_m , the following identity holds

$$|f_\alpha^{(q)}(t)|^2 = \sum_{n=0}^q B_{qn} |f^{(n)}(t)|^2 + 2 \sum_{0 \leq i \leq z \leq q} C_{qiz} \operatorname{Re}((-1)^{q-\frac{i+z}{2}} f^{(i)}(t) \overline{f^{(z)}(t)}). \quad (3)$$

where

$$B_{qn} = \binom{q}{n}^2 \alpha^{2(q-n)}, C_{qiz} = s_{qi} \binom{q}{i} \binom{q}{z} \alpha^{2q-z-i}, s_{qi} \in \{\pm 1\}, \quad (4)$$

and $\operatorname{Re}(\cdot)$ is the real part of $(\cdot)$.

Subsequently, we introduce an integral identity [7].

Identity 3. Suppose that $h(t)$, $\omega(t)$ and $\omega_p(t)$ are real-valued functions and all the following integrals exist. For any fixed but arbitrary $p \in \mathbb{N}$, $i = p - 2q \in \mathbb{N}^+$ and $r \in \{0, 1, 2, \dots, i-1\}$, the following identity holds

$$\int_{\mathbb{R}} \omega_p(t) h^{(i)}(t) dt = \sum_{r=0}^{i-1} (-1)^r \omega_p^{(r)}(t) h^{(i-r-1)}(t) + (-1)^i \int_{\mathbb{R}} \omega_p^{(i)}(t) h(t) dt, \quad (5)$$

$$\int_{\mathbb{R}} \omega_p(t) h^{(i)}(t) dt = (-1)^i \int_{\mathbb{R}} \omega_p^{(i)}(t) h(t) dt, \quad (6)$$

where

$$\omega_p(t) = (t - t_m)^p \omega(t), \text{ for any fixed but arbitrary constant } t_m \in \mathbb{R}. \quad (7)$$

Eq. (6) holds if the condition

$$\sum_{r=0}^{p-1} (-1)^r \lim_{|t| \rightarrow \infty} \omega_p^{(r)}(t) h^{(i-r-1)}(t) = 0, \quad (8)$$

holds.

We define the FT $\hat{f}(\xi)$ of the signal $f(t)$ as follows

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-2j\pi\xi t} f(t) dt, \quad (9)$$

and

$$f(t) = \int_{\mathbb{R}} e^{2j\pi\xi t} \hat{f}(\xi) d\xi, \quad (10)$$

then we introduce the PPR identity in the FT domain [7].

Identity 4. Suppose that $f(t)$, $f_\alpha^{(p)}(t)$, and $(\xi - \xi_m)^p \hat{f}(\xi)$ are in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, ξ_m are any fixed but arbitrary constants. For any fixed but arbitrary $p \in \mathbb{N}$, the following identity holds

$$\int_{\mathbb{R}} (\xi - \xi_m)^{2p} |\hat{f}(\xi)|^2 d\xi = \frac{1}{(2\pi)^{2p}} \int_{\mathbb{R}} |f_\alpha^{(p)}(t)|^2 dt. \quad (11)$$

2.3. The Heisenberg uncertainty principle related to the FT

This section introduces some Heisenberg uncertainty principles associated with the FT. First, we define

$$(\mu_{2p})_{\omega, |f(t)|^2} = \int_{\mathbb{R}} \omega^2(t) (t - t_m)^{2p} |f(t)|^2 dt, \quad (12)$$

and

$$(\mu_{2p})_{|\hat{f}(\xi)|^2} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |\hat{f}(\xi)|^2 d\xi, \quad (13)$$

then we review the classical $2p$ order HPW uncertainty principle [7].

Suppose that $f(t) \in L^2(\mathbb{R})$ and $\omega(t)$ is a real-valued weight function. Then the $2p$ order HPW uncertainty principle in the FT domain is given by

$$2^p \sqrt{(\mu_{2p})_{\omega, |f(t)|^2}} 2^p \sqrt{(\mu_{2p})_{|\hat{f}(\xi)|^2}} \geq \frac{1}{2\pi} \sqrt[2p]{|E_{p,f}|}, \quad (14)$$

where

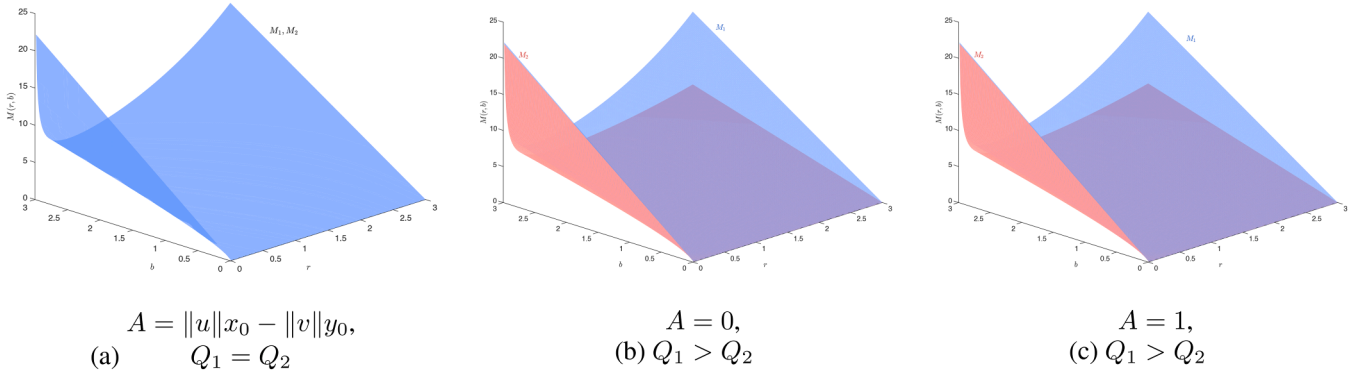
$$E_{p,f} = \sum_{q=0}^{\lfloor \frac{p}{2} \rfloor} D_q F_q, \quad 0 \leq q \leq \left\lfloor \frac{p}{2} \right\rfloor, \quad |F_q| < \infty, \quad (15)$$

$$D_q = (-1)^q \frac{p}{p-q} \binom{p-q}{q}, F_q = \sum_{n=0}^q B_{qn} I_{qn} + 2 \sum_{0 \leq i < z \leq q} C_{qiz} I_{qiz}. \quad (16)$$

Here

$$I_{qn} = (-1)^{q-2p} \int_{\mathbb{R}} \omega_p^{(p-2q)}(t) |f^{(n)}(t)|^2 dt, \quad (17)$$

$$I_{qiz} = (-1)^{q-2p} \int_{\mathbb{R}} \omega_p^{(p-2q)}(t) \operatorname{Re}((-1)^{q-\frac{i+z}{2}} f^{(i)}(t) \overline{f^{(z)}(t)}) dt, \quad (18)$$

Fig. 1. Numerical simulations for $p = 1$, $t_m = 0$ and $\tau = \xi_m = 0$.

B_{qn} and C_{qiz} are given by (4). In addition, we assume the following two conditions

$$\sum_{m=0}^{p-2q-1} (-1)^m \lim_{|t| \rightarrow \infty} \omega_p^{(m)}(t) \left(|f^{(m)}(t)|^2 \right)^{p-2q-m-1} = 0, \quad (19)$$

and

$$\sum_{m=0}^{p-2q-1} (-1)^m \lim_{|t| \rightarrow \infty} \omega_p^{(m)}(t) \left(\text{Re}(-1)^{q-\frac{i+\tau}{2}} f^{(i)}(t) \overline{f^{(z)}(t)} \right) = 0, \quad (20)$$

hold.

Next, let

$$(\mu_p)_{|f(t)|^2} = \int_{\mathbb{R}} |t - t_m|^p |f(t)|^2 dt, \quad (21)$$

and

$$(\mu_p)_{|\hat{f}(\xi)|^2} = \int_{\mathbb{R}} |\xi - \xi_m|^p |\hat{f}(\xi)|^2 d\xi, \quad (22)$$

then we introduce the HW uncertainty principle [13].

Suppose that $E_{|f(t)|^2} = \int_{\mathbb{R}} |f(t)|^2 dt$ and $p \geq 2$, then the HW uncertainty principle in the FT domain is as

$$\sqrt[p]{(\mu_p)_{|f(t)|^2}} \sqrt[p]{(\mu_p)_{|\hat{f}(\xi)|^2}} \geq \frac{1}{4\pi} \sqrt[p]{E_{|f(t)|^2}^2}. \quad (23)$$

Then we present the SHW uncertainty principle [14].

Suppose that $f(t) \in L^2(\mathbb{R})$, and for any fixed but arbitrary $p \in \mathbb{N}$, the SHW uncertainty principle associated with the FT holds as follows

$$\sqrt[p]{(\mu_{2p})_{\omega, |f(t)|^2}} \sqrt[p]{(\mu_{2p})_{|\hat{f}(\xi)|^2}} \geq \frac{1}{2\pi \sqrt[p]{2}} \sqrt[p]{|E_{p,f}^*|}, \quad (24)$$

where

$$|E_{p,f}^*| = \sqrt{E_{p,f}^2 + 4A^*{}^2}, \quad (25)$$

and $(\mu_{2p})_{\omega, |f(t)|^2}$ is given by (12), $(\mu_{2p})_{|\hat{f}(\xi)|^2}$ is given by (13), $|E_{p,f}^*|$ is given by (15). The equality in (24) holds when

$$A^* = \|u^*(t)\|x_0^* - \|v^*(t)\|y_0^*, \quad (26)$$

where $u^*(t) = \omega(t)(t - t_m)^p f_a(t)$, $v^*(t) = f_a^{(p)}(t)$ and $x_0^* = \int_{\mathbb{R}} |v^*(t)| |h(t)| dt$, $y_0^* = \int_{\mathbb{R}} |u^*(t)| |h(t)| dt$, $\|h(t)\|^2 = \int_{\mathbb{R}} |h(t)|^2 dt = 1$, t_m are any fixed but arbitrary constants.

3. Main results

This section derives the main theorems of this paper, including the PPR identity, the $2p$ order HPW uncertainty principle, the HW uncertainty principle, and the SHW uncertainty principle in the OLCT domain.

3.1. The PPR identity of the OLCT

To derive the $2p$ order Heisenberg uncertainty principles in the OLCT domain, we first derived the PPR identity, which can be represented as the following Theorem 1.

Theorem 1. Suppose that $f(t)$, $O_f^J(\xi)$, $(\xi - \xi_m)^p O_f^J(\xi)$ belong to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, ξ_m are any fixed but arbitrary constants. For any fixed but arbitrary $p \in \mathbb{N} = \{0, 1, 2, \dots\}$, we have

$$\int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi = b^{2p} \int_{\mathbb{R}} |g_{\beta}^{(p)}(t)|^2 dt, \quad (27)$$

where $g_{\beta}^{(p)}(t) = \frac{d^p}{dt^p} e^{-j\beta t} e^{j\frac{a}{2b}t^2} f(t)$ with $\beta = 2\pi\sigma_m$, $\sigma_m = \frac{\xi_m - \tau}{2\pi b}$.

Proof. Regarding the left-hand side of (27), it can be rewritten as

$$\begin{aligned} & \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi \\ &= \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \frac{1}{\sqrt{j2\pi b}} \int_{\mathbb{R}} f(t) e^{j[\frac{a}{2b}t^2 - \frac{1}{b}t(\xi - \tau) - \frac{1}{b}\xi(d\tau - b\eta) + \frac{d}{2b}(\xi^2 + \tau^2)]} dt \right|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \int_{\mathbb{R}} f(t) e^{j[\frac{a}{2b}t^2 - \frac{(\xi - \tau)t}{b}]} dt \right|^2 d\xi. \end{aligned}$$

Since $g(t) = f(t) e^{j\frac{a}{2b}t^2}$, then

$$\begin{aligned} & \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \int_{\mathbb{R}} g(t) e^{-j\frac{(\xi - \tau)t}{b}} dt \right|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (\xi - \xi_m)^{2p} \left| \hat{g}\left(\frac{\xi - \tau}{2\pi b}\right) \right|^2 d\xi \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} [\xi - \tau - (\xi_m - \tau)]^{2p} \left| \hat{g}\left(\frac{\xi - \tau}{2\pi b}\right) \right|^2 d(\xi - \tau) \\ &= \frac{1}{2\pi b} \int_{\mathbb{R}} (2\pi b)^{2p} (\sigma - \sigma_m)^{2p} |\hat{g}(\sigma)|^2 d(2\pi b\sigma) \\ &= (2\pi b)^{2p} \int_{\mathbb{R}} (\sigma - \sigma_m)^{2p} |\hat{g}(\sigma)|^2 d\sigma. \end{aligned} \quad (28)$$

Using (11), we obtain

$$\int_{\mathbb{R}} (\sigma - \sigma_m)^{2p} |\hat{g}(\sigma)|^2 d\sigma = \frac{1}{(2\pi)^{2p}} \int_{\mathbb{R}} |g_{\beta}^{(p)}(t)|^2 dt. \quad (29)$$

Substituting (29) into (28) completes the proof. $\square$

3.2. The $2p$ order HPW uncertainty principle of the OLCT

We define

$$(\mu_{2p})_{|O_f^J(\xi)|^2} = \int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi, \quad (30)$$

to establish the $2p$ order HPW uncertainty principle in the OLCT domain as following Theorem 2.

Theorem 2. Suppose that $f(t)$, $O_f^J(\xi)$, $(\xi - \xi_m)^p O_f^J(\xi)$ belong to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Let $\omega(t)$ be a real weight function. For any fixed but arbitrary $p \in \mathbb{N}$, the $2p$ order HPW uncertainty principle in the OLCT domain is as follows

$${}^{2p}\sqrt{(\mu_{2p})_{\omega, |f(t)|^2}} {}^{2p}\sqrt{(\mu_{2p})_{|O_f^J(\xi)|^2}} \geq \frac{|b|}{\sqrt[2]{2}} \sqrt[2]{E_{p,f}}, \quad (31)$$

where $(\mu_{2p})_{\omega, |f(t)|^2}$ is given by (12) and $E_{p,f}$ is given by (15).

Remark 1. From Theorem 2, we observe that

$$(\mu_{2p})_{|O_f^J(\xi)|^2} = b^{2p} \int_{\mathbb{R}} \left| \frac{d^{2p}}{dt^{2p}} \left(e^{-j\frac{\xi m}{b}t} e^{j\frac{a}{2b}t^2} f(t) \right) \right|^2 dt,$$

which explicitly depends on the offset parameter τ in the OLCT domain. Therefore, in contrast to the HPW uncertainty principle in the LCT domain, the presence of the offset parameter τ introduces additional modulation flexibility and provides a more generalized formulation for HPW uncertainty principle.

Corollary 1. For $p = 1$, the second-order HPW uncertainty principle in the OLCT domain can be obtained as follows

$$\sqrt{(\mu_2)_{\omega, |f(t)|^2}} \sqrt{(\mu_2)_{|O_f^J(\xi)|^2}} \geq \frac{|b|}{2} \sqrt{E_{1,f}}, \quad (32)$$

where

$$E_{1,f} = \int_{\mathbb{R}} [(t - t_m)\omega(t)]' |f(t)|^2 dt. \quad (33)$$

Equality holds in (32) if and only if

$$f(t) = c_0 e^{-c_p(t-t_m)^2} e^{j(\frac{\xi m}{b}t - \frac{a}{2b}t^2)}, \quad (34)$$

where $c_0, t_m, \xi_m \in \mathbb{R}$ and $c_p \in \mathbb{R}^+$.

Corollary 2. For $p = 1$, assuming $\omega(t) = 1$ and $\|f(t)\|_2^2 = \int_{\mathbb{R}} |f(t)|^2 dt = E_{|f|^2}$, the second-order HPW uncertainty principle in the OLCT domain can be derived as follows

$$(\mu_2)_{1, |f(t)|^2} (\mu_2)_{|O_f^J(\xi)|^2} \geq \frac{b^2}{4} \|f\|_2^4. \quad (35)$$

When $\|f(t)\|_2 = 1$ and $\xi_m = \int_{\mathbb{R}} \xi |O_f^J(\xi)|^2 d\xi$, Eq. (35) reduces to the Heisenberg uncertainty principle derived in Stern [32].

Corollary 3. When the parameters of the OLCT are set to $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, Theorem 2 reduces to the $2p$ order uncertainty principle in the FT domain given in (14).

3.3. The HW uncertainty principle of the OLCT

Let

$$(\mu_p)_{|O_f^J(\xi)|^2} = \int_{\mathbb{R}} |\xi - \xi_m|^p |O_f^J(\xi)|^2 d\xi, \quad (36)$$

then we can obtain the following Theorem 3.

Theorem 3. Suppose that $f(t)$, $O_f^J(\xi)$, $(\xi - \xi_m)^p O_f^J(\xi)$ belong to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. For $p \geq 2$ and any fixed but arbitrary real constants t_m, ξ_m , the HW uncertainty principle of the OLCT holds as follows

$$\sqrt[2]{(\mu_p)_{|f(t)|^2}} \sqrt[2]{(\mu_p)_{|O_f^J(\xi)|^2}} \geq \frac{|b|}{2} \sqrt[2]{E_{|f(t)|^2}^2}, \quad (37)$$

where $(\mu_p)_{|f(t)|^2}$ is given by (21) and $E_{|f(t)|^2} = \int_{\mathbb{R}} |f(t)|^2 dt$.

Proof. Firstly,

$$\begin{aligned} (\mu_p)_{|f(t)|^2}^{\frac{2}{p}} E_{|f(t)|^2}^{1-\frac{2}{p}} &= \left(\int_{\mathbb{R}} |t - t_m|^p |f(t)|^2 dt \right)^{\frac{2}{p}} \left(\int_{\mathbb{R}} |f(t)|^2 dt \right)^{1-\frac{2}{p}} \\ &= \left[\int_{\mathbb{R}} \left(|t - t_m|^2 |f(t)|^{\frac{4}{p}} \right)^{\frac{p}{2}} dt \right]^{\frac{2}{p}} \left[\int_{\mathbb{R}} \left(|f(t)|^{2(1-\frac{2}{p})} \right)^{1/(1-\frac{2}{p})} dt \right]^{1-\frac{2}{p}}. \end{aligned} \quad (38)$$

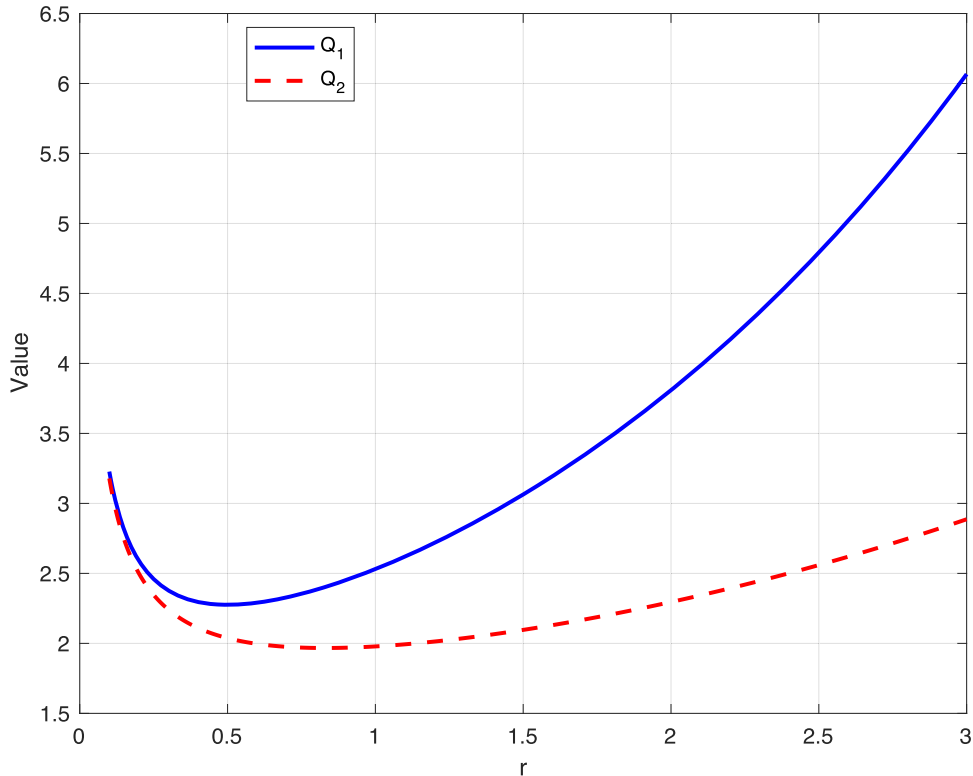
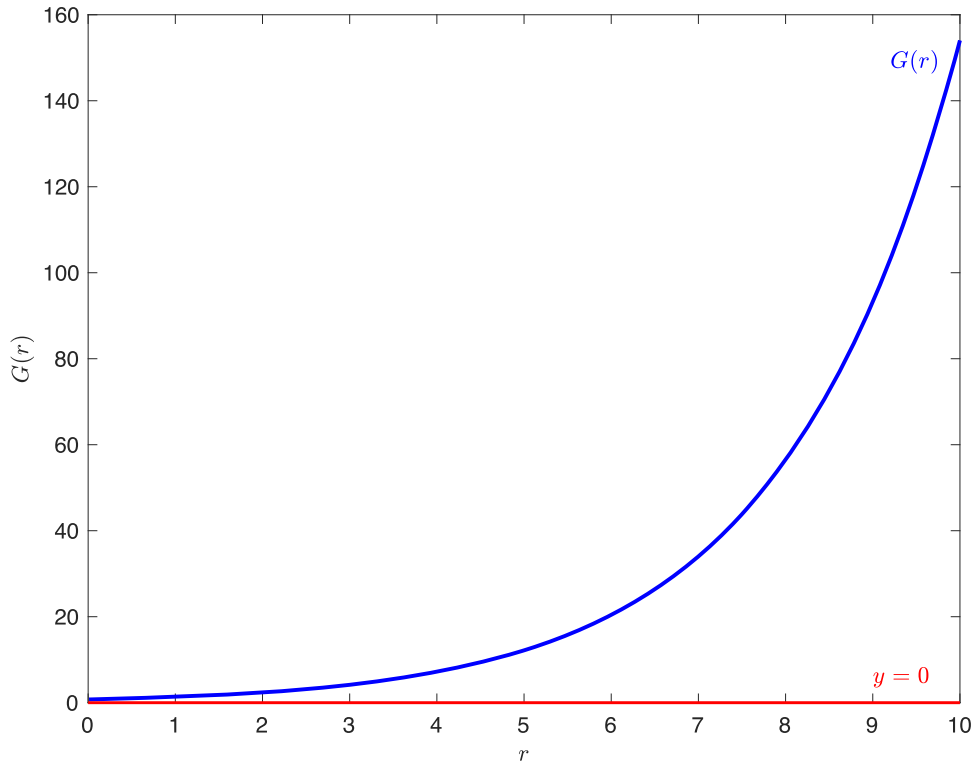


Fig. 2. Numerical simulations for $p = 1$, $t_m = 0$ and $\tau = \xi_m = 0$.

Fig. 3. The plot of $G(r)$.

Then applying the Hölder inequality to (38), we obtain

$$\begin{aligned} (\mu_p)_{|f(t)|^2}^{\frac{2}{p}} E_{|f(t)|^2}^{1-\frac{2}{p}} &\geq \int_{\mathbb{R}} \left((t-t_m)^2 |f(t)|^{\frac{4}{p}} \right) \left(|f(t)|^{2(1-\frac{2}{p})} \right) dt \\ &= \int_{\mathbb{R}} (t-t_m)^2 |f(t)|^2 dt \\ &= (\mu_2)_{|f(t)|^2} \triangleq \phi_{|f(t)|^2}^2, \end{aligned}$$

so

$$(\mu_p)_{|f(t)|^2}^{\frac{1}{p}} \geq \phi_{|f(t)|^2} / E_{|f(t)|^2}^{\frac{1}{2}-\frac{1}{p}}. \quad (39)$$

Equality holds in (39) if and only if

$$|t-t_m|^p E_{|f(t)|^2} = (\mu_p)_{|f(t)|^2}.$$

Similarly,

$$\begin{aligned} (\mu_p)_{|O_f^J(\xi)|^2}^{\frac{2}{p}} E_{|O_f^J(\xi)|^2}^{1-\frac{2}{p}} &= \left(\int_{\mathbb{R}} |\xi - \xi_m|^p |O_f^J(\xi)|^2 d\xi \right)^{\frac{2}{p}} \left(\int_{\mathbb{R}} |O_f^J(\xi)|^2 d\xi \right)^{1-\frac{2}{p}} \\ &\geq \int_{\mathbb{R}} (\xi - \xi_m)^2 |O_f^J(\xi)|^2 d\xi \\ &= (\mu_2)_{|O_f^J(\xi)|^2} \triangleq \phi_{|O_f^J(\xi)|^2}^2, \end{aligned}$$

so

$$(\mu_p)_{|O_f^J(\xi)|^2}^{\frac{1}{p}} \geq \phi_{|O_f^J(\xi)|^2} / E_{|O_f^J(\xi)|^2}^{\frac{1}{2}-\frac{1}{p}}. \quad (40)$$

Equality holds in (40) if and only if

$$|\xi - \xi_m|^p E_{|O_f^J(\xi)|^2} = (\mu_p)_{|O_f^J(\xi)|^2}.$$

By the Parseval property of the OLCT [33], we obtain

$$E_{|f(t)|^2} = E_{|O_f^J(\xi)|^2}, \quad (41)$$

then we find

$$(\mu_p)_{|f(t)|^2}^{\frac{1}{p}} (\mu_p)_{|O_f^J(\xi)|^2}^{\frac{1}{p}} \geq \phi_{|f(t)|^2} \phi_{|O_f^J(\xi)|^2} / E_{|f(t)|^2}^{1-\frac{2}{p}}. \quad (42)$$

From (35), we have

$$\phi_{|f(t)|^2}^2 \phi_{|O_f^J(\xi)|^2}^2 \geq \frac{b^2}{4} E_{|f(t)|^2}^2. \quad (43)$$

Substituting Eq. (43) into (42) completes the proof of Theorem 3. $\square$

Corollary 4. When the parameters of the OLCT are set to $J = \begin{bmatrix} a & b & 0 \\ c & d & 0 \end{bmatrix}$, the HW uncertainty principle in the LCT domain can be obtained.

3.4. The SHW uncertainty principle of the OLCT

This section builds upon the $2p$ order HPW uncertainty principle in the OLCT domain and presents the SHW uncertainty principle, which provides a more precise lower bound.

Theorem 4. Suppose that $f(t)$, $O_f^J(\xi)$, $(\xi - \xi_m)^p O_f^J(\xi)$ belong to $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. For any fixed but arbitrary $p \in \mathbb{N}$, the SHW uncertainty principle of the OLCT holds as follows

$${}^{2p}\sqrt{(\mu_{2p})_{\omega, |f(t)|^2}} {}^{2p}\sqrt{(\mu_{2p})_{|O_f^J(\xi)|^2}} \geq \frac{|b|}{\sqrt{2}} \sqrt{|E_{p,f}^*|}. \quad (44)$$

The equality in (44) holds when

$$A = \|u(t)\|x_0 - \|v(t)\|y_0, \quad (45)$$

where $u(t) = \omega(t)(t-t_m)^p g_\beta(t)$, $v(t) = g_\beta^{(p)}(t)$ and $x_0 = \int_{\mathbb{R}} |v(t)||h(t)|dt$, $y_0 = \int_{\mathbb{R}} |u(t)||h(t)|dt$, $\|h(t)\|^2 = \int_{\mathbb{R}} |h(t)|^2 dt = 1$.

Proof. In fact, from (27) in Theorem 1, we obtain

$$\begin{aligned} M_p^* &= M_p - b^{2p} A^2 \\ &= (\mu_{2p})_{\omega, |f(t)|^2} (\mu_{2p})_{|O_f^J(\xi)|^2} - b^{2p} A^2 \\ &= \left(\int_{\mathbb{R}} \omega^2(t)(t-t_m)^{2p} |f(t)|^2 dt \right) \cdot \left(\int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi \right) - b^{2p} A^2 \end{aligned}$$

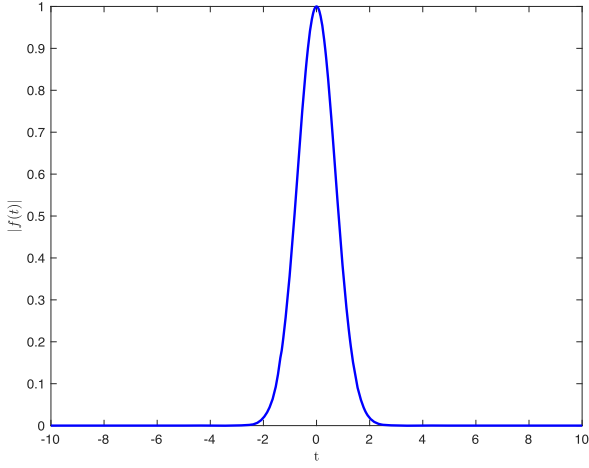
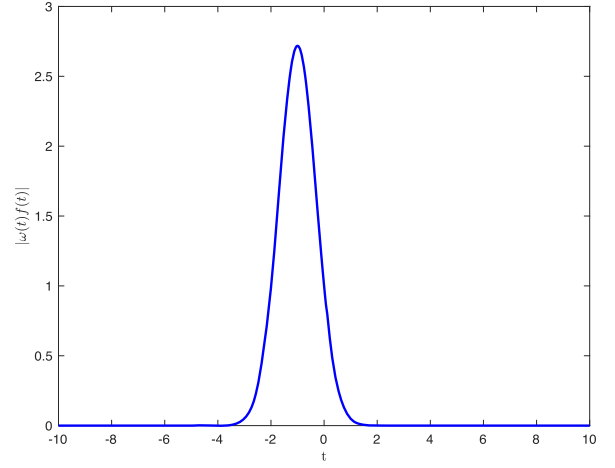
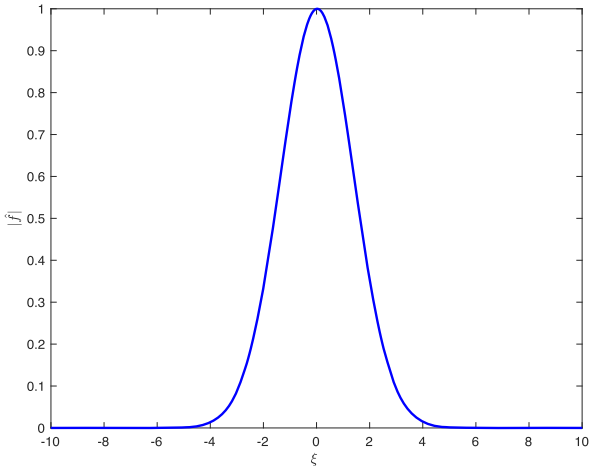
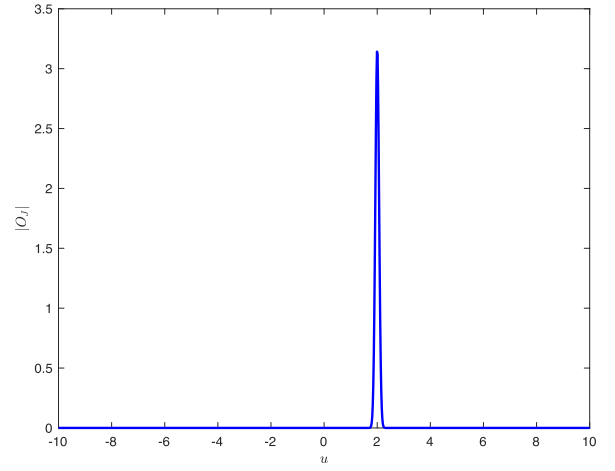
(a) Energy density of $f(t)$ (b) Energy density of weighted $f(t)$ (c) Energy density of $\hat{f}(\xi)$ (d) Energy density of $O_J(\xi)$

Fig. 4. The energy of a signal that can achieve the lower bound.

$$\begin{aligned}
 &= b^{2p} \left[\left(\int_{\mathbb{R}} \omega^2(t) (t - t_m)^{2p} |g_\beta(t)|^2 dt \right) \left(\int_{\mathbb{R}} |g_\beta^{(p)}(t)|^2 dt \right) - A^2 \right] \\
 &= b^{2p} (\|u\|^2 \|v\|^2 - A^2).
 \end{aligned} \quad (46)$$

From the positive definiteness of the following Gram determinant, we obtain

$$\begin{aligned}
 0 &\leq \begin{pmatrix} \|u\|^2 & (|u|, |v|) & y_0 \\ (|u|, |v|) & \|v\|^2 & x_0 \\ y_0 & x_0 & 1 \end{pmatrix} \\
 &= \|u\|^2 \|v\|^2 - (|u|, |v|)^2 - [\|u\|^2 x_0^2 - 2(|u|, |v|) x_0 y_0 + \|v\|^2 y_0^2] \\
 &\leq \|u\|^2 \|v\|^2 - (|u|, |v|)^2 - A^2,
 \end{aligned}$$

then we find

$$\begin{aligned}
 M_p^* &\geq b^{2p} (|u|, |v|)^2 \\
 &= b^{2p} \left(\int_{\mathbb{R}} |u| |v| dt \right)^2 \\
 &= b^{2p} \left(\int_{\mathbb{R}} |\omega_p(t) g_\beta(t) g_\beta^{(p)}(t)| dt \right)^2.
 \end{aligned} \quad (47)$$

The right-hand sides of Eqs. (A.3) and (47) are exactly the same, so we obtain

$$M_p^* \geq \frac{b^{2p}}{4} E_{p,f}^2. \quad (48)$$

Substituting Eqs. (48) into (46), we have

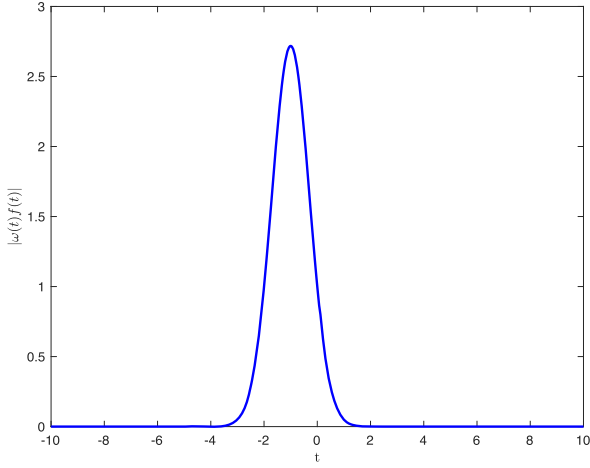
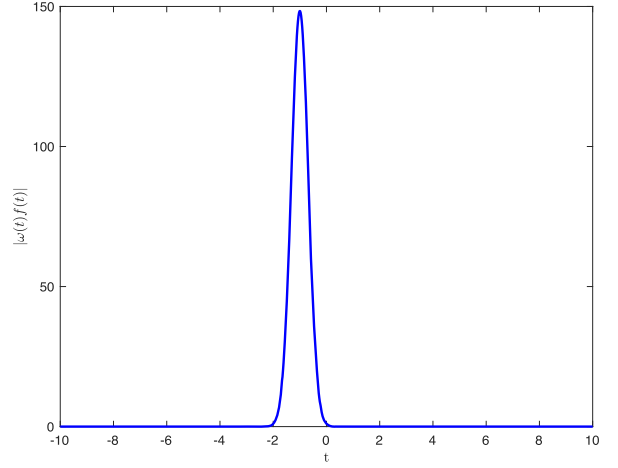
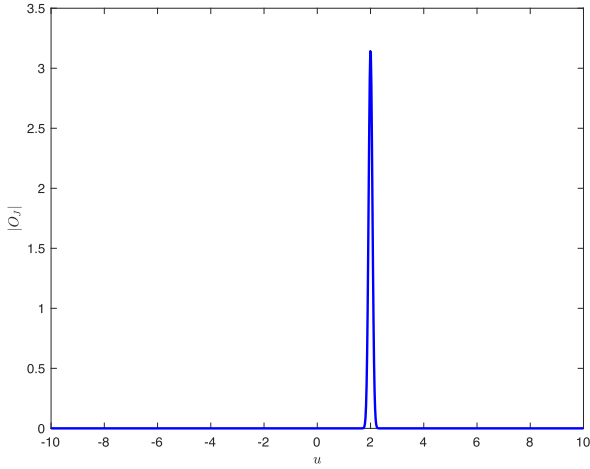
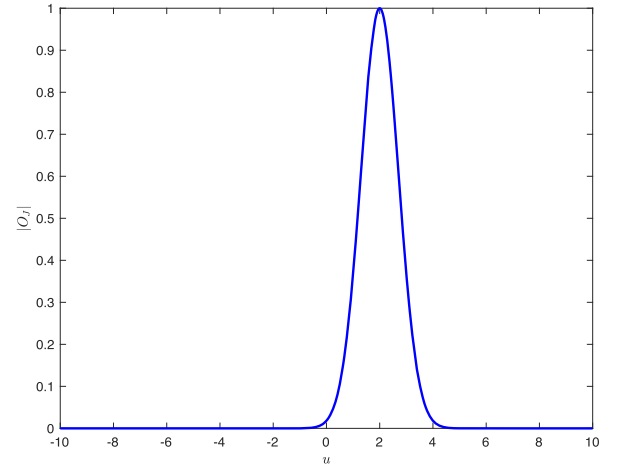
$$\begin{aligned}
 M_p &\geq \frac{b^{2p}}{4} E_{p,f}^2 + b^{2p} A^2 \\
 &= \frac{b^{2p}}{4} (E_{p,f}^2 + 4A^2).
 \end{aligned} \quad (49)$$

So the $2p$ order SHW uncertainty principle of the OLCT is as follows

$$\sqrt[p]{M_p} \geq \frac{|b|}{\sqrt[p]{2}} \sqrt[p]{E_{p,f}^*},$$

where $|E_{p,f}^*| = \sqrt{E_{p,f}^2 + 4A^2}$. Hence, Theorem 4 is proved. $\square$

Theorems 2 and 4 provide two lower bounds for the HPW uncertainty principle of the OLCT. From a mathematical standpoint, a larger value on the right side of the inequality indicates a more precise and reliable estimate of the left side. Therefore, it is crucial to study the sharpened version of the uncertainty principle. From a practical signal

(a) $\omega(t) = e^{-2t}$ (b) $\omega(t) = e^{-10t}$ (c) $a = 0.6, b = 0.05$ (d) $a = 6, b = 0.5$ **Fig. 5.** Influence of the weight function ω and parameters a and b on the energy distribution of the given signal.

processing perspective, a smaller lower bound implies that higher time-frequency concentration is theoretically permissible in time-frequency analysis, thereby characterizing the ideal theoretical limit. However, it remains necessary to derive the sharpened version of the uncertainty principle in the OLCT domain with larger lower bound. The reason is that a smaller theoretical bound merely reflects the potential limit under ideal conditions, whereas actual signals often fail to attain this level of concentration. In contrast, the sharpened lower bound, although numerically larger, provides a more precise estimate that is closer to the true achievable limit.

4. Simulations

This section validates the results of this paper through numerical experiments. Taking [Theorem 4](#) as an example, the superiority of the numerical results is demonstrated and the factors influencing signal concentration in the OLCT domain are analyzed.

Example 1. Consider a signal $f(t) = e^{-\frac{r}{2}t^2} e^{-j\frac{a}{2b}t^2}$ and a real weight function $\omega(t) = e^{-rt}$ with $r > 0$.

We let $Q_1 = \sqrt[2p]{(\mu_{2p})_{\omega, |f(t)|^2}} \sqrt[2p]{(\mu_{2p})_{|Q_f(\xi)|^2}}$ and $Q_2 = \frac{b}{\sqrt[2p]{2}} \sqrt[2p]{E_{p,f}^*}$. For different values of A , we compute Q_1 and Q_2 respectively and then plot the corresponding graphs in [Fig. 1](#).

[Fig. 1](#) illustrates that when $A = \|u\|_{x_0} - \|v\|_{y_0}$, we obtain $Q_1 = Q_2$. In this case, the product of the time resolution and the frequency resolution in the OLCT domain can reach its minimum value. In other words, the sharpened HPW uncertainty principle can achieve the theoretical lower bound.

For different values of r , we compute Q_1 and Q_2 with $b = 1$ and $A = 1$, then plot the corresponding graph in [Fig. 2](#).

From [Fig. 2](#), it can be observed that $Q_1 > Q_2$ holds for all $r > 0$, thereby confirming the validity of [Theorem 4](#).

Furthermore, we compare the lower bounds provided by different $2p$ order uncertainty principles in [Table 1](#). The signal in [Example 1](#) is identical to that in [Feng et al. \[25\]](#). Hence, we compare our result with theirs.

Table 1

Comparison of the lower bounds of different uncertainty principles.

Uncertainty principles	The lower bound in Example 1
The $2p$ order uncertainty principle derived in this paper	$\frac{b^2}{2} \pi e^r \left(\frac{1}{2r} + 1 \right)$
The $2p$ order uncertainty principle derived in Feng et al. [25]	$\frac{b^2}{4} \frac{\pi}{r} e^{\frac{r}{2}} \left(1 + \frac{r}{2} \right)^2$

Compared with the result in Feng et al. [25], it can be concluded that for $r > 0$, the inequality

$$\frac{b^2}{2} \pi e^r \left(\frac{1}{2r} + 1 \right) > \frac{b^2}{4} \frac{\pi}{r} e^{\frac{r}{2}} \left(1 + \frac{r}{2} \right)^2, \quad (50)$$

holds. This indicates that the SHW uncertainty principle provides a tighter lower bound. Then we prove the (50) holds universally. The difference between the two sides of the inequality (50) can be explicitly expressed as

$$\frac{b^2}{2} \pi e^r \left(\frac{1}{2r} + 1 \right) - \frac{b^2}{4} \frac{\pi}{r} e^{\frac{r}{2}} \left(1 + \frac{r}{2} \right)^2 = \frac{b^2}{2} \pi e^{\frac{r}{2}} \left[e^{\frac{r}{2}} \left(\frac{1}{2r} + 1 \right) - \frac{1}{2r} \left(1 + \frac{r}{2} \right)^2 \right].$$

Let $G(r) = e^{\frac{r}{2}} \left(\frac{1}{2r} + 1 \right) - \frac{1}{2r} \left(1 + \frac{r}{2} \right)^2$ and its graph is shown in [Fig. 3](#).

[Fig. 3](#) clearly shows that $G(r) > 0$ for all $r > 0$, confirming the universal validity of inequality (50).

Example 2. Consider the signal in [Example 1](#) with $r = 2$, then $f(t) = e^{-t^2} e^{-j6t^2}$ and $\omega(t) = e^{-2t}$. We set the parameters as $a = 0.6$, $b = 0.05$, $c = 0.5$, $d = 0.4$, $\tau = 2$, $\eta = 0$, $t_m = 0$, $\xi_m = 0$.

Compared with the FT, the OLCT allows for better energy concentration of signals through appropriate parameter selection. The following figure illustrates the energy distribution of the original signal, the weighted signal, the signal in the FT domain, and the signal in the OLCT domain.

As shown in [Fig. 4](#), when appropriate parameters are chosen for the OLCT, the signal energy exhibits greater concentration in the OLCT domain than in the conventional Fourier domain. And it is easy to calculate that

$$\sqrt{(\mu_2)_{\omega, |f(t)|^2}} \sqrt{(\mu_2)_{|O_f^J(\xi)|^2}} = 1.904493221525881,$$

$$\frac{|b|}{2} |E_{p,f}^*| = 1.904493221525881.$$

The equality $\sqrt{(\mu_2)_{\omega, |f(t)|^2}} \sqrt{(\mu_2)_{|O_f^J(\xi)|^2}} = \frac{|b|}{2} |E_{p,f}^*|$ validates the conclusion of [Theorem 4](#) for the case when $p = 1$.

Example 3. Consider signals in [Example 1](#) with $r = 2$ and $r = 10$, respectively. Then the signals are given by $f(t) = e^{-t^2} e^{-j6t^2}$ and $f(t) = e^{-5t^2} e^{-j6t^2}$. And the real weight functions are given by $\omega(t) = e^{-2t}$ and $\omega(t) = e^{-10t}$. Two settings of the parameters are considered: one with $a = 0.6$, $b = 0.05$ and the other with $a = 6$, $b = 0.5$, while keeping $c = 0.5$, $d = 0.4$, $\tau = 2$, $\eta = 0$ fixed.

To illustrate the influence of the weight function $\omega(t)$ and the OLCT parameters on the energy concentration of the signal, we present the energy distributions under different weight functions and various parameter settings.

[Fig. 5](#) demonstrates that the concentration of the signal's energy is dependent on the choice of the weight function ω and the parameters a and b associated with the OLCT. In particular, a faster decay of the weight function ω leads to a more pronounced localization of the signal's energy.

5. Conclusion

The uncertainty principle plays a pivotal role in signal processing and its applications. The offset linear canonical transform (OLCT) provides a more flexible and general approach due to its offset parameters.

In this paper, the Plancherel-Parseval-Rayleigh identity in the OLCT domain is derived. Based on this identity, the $2p$ order Heisenberg-Pauli-Weyl uncertainty principle of the OLCT is established. Subsequently, the Heisenberg-Weyl uncertainty principle in the OLCT domain is deduced. The sharpened Heisenberg-Weyl uncertainty principle is then proposed and further validated through numerical simulations. The results of this work provide theoretical advancements and provide valuable insights for practical applications in time-frequency analysis within the OLCT domain.

CRedit authorship contribution statement

Jia-Yin Peng: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft; **Bing-Zhao Li:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by grants from the [National Natural Science Foundation of China](#) [No. 62171041].

Appendix A. Proof of [Theorem 2](#)

To prove [Theorem 2](#), we set

$$\begin{aligned} M_p &= (\mu_{2p})_{\omega, |f(t)|^2} (\mu_{2p})_{|O_f^J(\xi)|^2} \\ &= \left(\int_{\mathbb{R}} \omega^2(t) (t - t_m)^{2p} |f(t)|^2 dt \right) \cdot \left(\int_{\mathbb{R}} (\xi - \xi_m)^{2p} |O_f^J(\xi)|^2 d\xi \right). \end{aligned} \quad (A.1)$$

Since $g_\beta(t) = e^{-j\beta t} e^{\frac{a}{2b} t^2} f(t)$ and $|g_\beta(t)|^2 = |f(t)|^2$, then from (27) and (A.1), we obtain

$$M_p = b^{2p} \left(\int_{\mathbb{R}} \omega^2(t) (t - t_m)^{2p} |g_\beta(t)|^2 dt \right) \left(\int_{\mathbb{R}} |g_\beta^{(p)}(t)|^2 dt \right). \quad (A.2)$$

Based on (A.2), (7) and applying the Cauchy-Schwarz inequality, we get

$$M_p \geq b^{2p} \left(\int_{\mathbb{R}} |\omega_p(t) g_\beta(t) g_\beta^{(p)}(t)| dt \right)^2. \quad (A.3)$$

Applying the complex inequality to (A.3), we have

$$M_p \geq \frac{b^{2p}}{4} \left(\int_{\mathbb{R}} \omega_p(t) \left(g_\beta(t) \overline{g_\beta^{(p)}(t)} + \overline{g_\beta(t)} g_\beta^{(p)}(t) \right) dt \right)^2. \quad (A.4)$$

Using the differential identity (2) on (A.4), we obtain

$$M_p \geq \frac{b^{2p}}{4} \left(\int_{\mathbb{R}} \omega_p(t) \left(\sum_{q=0}^{\left\lfloor \frac{p}{2} \right\rfloor} D_q \frac{d^{p-2q}}{dt^{p-2q}} |g_\beta^{(q)}(t)|^2 \right) dt \right)^2. \quad (A.5)$$

Applying the Lagrange type differential identity (3) to $|g_\beta^{(q)}(t)|^2$, we get

$$|g_\beta^{(q)}(t)|^2 = \sum_{n=0}^q B_{qn} |g^{(n)}(t)|^2 + 2 \sum_{0 \leq i < z \leq q} C_{qiz} \operatorname{Re}((-1)^{q-\frac{i+z}{2}} g^{(i)}(t) \overline{g^{(z)}(t)}). \quad (A.6)$$

From the integral identity (6) and (A.6), we find

$$\int_{\mathbb{R}} \omega_p(t) \frac{d^{p-2q}}{dt^{p-2q}} |g^{(n)}(t)|^2 dt = (-1)^{p-2q} \int_{\mathbb{R}} \omega_p^{p-2q}(t) |g^{(n)}(t)|^2 dt = I_{qn}, \quad (A.7)$$

$$\begin{aligned}
& \int_{\mathbb{R}} \omega_p(t) \frac{d^{p-2q}}{dt^{p-2q}} \operatorname{Re}((-1)^{q-\frac{i+z}{2}} g^{(i)}(t) \overline{g^{(z)}(t)}) dt \\
&= (-1)^{p-2q} \int_{\mathbb{R}} \omega_p^{p-2q}(t) \operatorname{Re}((-1)^{q-\frac{i+z}{2}} g^{(i)}(t) \overline{g^{(z)}(t)}) dt \\
&= I_{qiz}.
\end{aligned} \tag{A.8}$$

Taking (A.6)–(A.8) into (A.5), we obtain

$$\begin{aligned}
M_p &\geq \frac{b^{2p}}{4} \left(\sum_{q=0}^{\left\lfloor \frac{p}{2} \right\rfloor} D_q \left(\sum_{l=0}^q B_{qn} I_{qn} + 2 \sum_{0 \leq i < z \leq q} C_{qiz} I_{qiz} \right) \right)^2 \\
&= \frac{b^{2p}}{4} (E_{p,f})^2.
\end{aligned} \tag{A.9}$$

This completes the proof of Theorem 2.

References

- [1] W. Heisenberg, Über den anschaulichen inhalt der quantentheoretischen kinematik und mechanik, *Z. Physik* 43 (3) (1927) 172–198.
- [2] H. Weyl, Quantenmechanik und gruppentheorie, *Z. Physik* 46 (1) (1927) 1–46.
- [3] E.H. Kennard, Zur quantenmechanik einfacher bewegungstypen, *Z. Physik* 44 (4) (1927) 326–352.
- [4] C. Gasquet, P. Witomski, *Fourier Analysis and Applications: Filtering, Numerical Computation, Wavelets*, Springer, New York, 2013.
- [5] N.R. Bracewell, *The Fourier Transform and its Applications*, McGraw Hill New York, 2000.
- [6] C. Wang, J.P. Yao, Fourier transform ultrashort optical pulse shaping using a single chirped fiber Bragg grating, *IEEE Photon. Technol. Lett* 21 (19) (2009) 1375–1377.
- [7] J.M. Rassias, On the Heisenberg-Pauli-Weyl inequality, *J. Inequal. Pure Appl. Math* 5 (1) (2004) 1–151.
- [8] V.C. Chen, H. Ling, *Time-Frequency Transforms for Radar Imaging and Signal Analysis*, 2002. Artech house.
- [9] G.L. Xu, X.T. Wang, L.J. Zhou, L.M. Shao, Y.L. Liu, X.G. Xu, Study on the mathematical problems of generalized uncertainty principles, *Adv. Appl. Math.* 5 (3) (2016) 536–559.
- [10] W.B. Gao, B.Z. Li, Uncertainty principles for the short-time linear canonical transform of complex signals, *Digit. Signal Process* 111 (2021) 102953.
- [11] Z.C. Zhang, Uncertainty principle for linear canonical transform using matrix decomposition of absolute spread matrix, *Digit. Signal Process*, 89, 2019.
- [12] Z.C. Zhang, Tighter uncertainty principles for linear canonical transform in terms of matrix decomposition, *Digit. Signal Process* 69 (2017) 70–85.
- [13] J.M. Rassias, On the Heisenberg-Weyl inequality, *J. Inequ. Pure Appl. Math.* 6 (1) (2005) 1–18.
- [14] J.M. Rassias, On the sharpened Heisenberg-Weyl inequality, *J. Inequ. Pure Appl. Math.* 7 (3) (2006) 1–8.
- [15] V. Namias, The fractional order Fourier transform and its application to quantum mechanics, *IMA J. Appl. Math.* 25 (3) (1980) 241–265.
- [16] R. Tao, B. Deng, Y. Wang, Research progress of the fractional Fourier transform in signal processing, *Sci. China Ser. F* 49 (2006) 1–25.
- [17] N. Kaur, B. Gupta, A.K. Verma, Multidimensional fractional wavelet transforms and uncertainty principles, *J. Comput. Appl. Math.* 430 (2023) 115250.
- [18] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, *Linear canonical transforms*, Springer Ser. Opt. Sci. 198 (2016) 453.
- [19] J.Y. Peng, J.Y. Chen, B.Z. Li, A novel kind of WVD associated with the linear canonical transform, Technical Report, APSIPA ASC, 2024.
- [20] S.C. Pei, J.J. Ding, Eigenfunctions of the offset Fourier, fractional Fourier, and linear canonical transforms, *J. Opt. Soc. Am A* 20 (3) (2003) 522–532.
- [21] S.C. Pei, J.J. Ding, Eigenfunctions of Fourier and fractional Fourier transforms with complex offsets and parameters, *IEEE Trans. Circuit. Syst. I, Reg. Paper.* 54 (7) (2007) 1599–1611.
- [22] G.L. Xu, X.T. Wang, X.G. Xu, The logarithmic, Heisenberg's and shorttime uncertainty principles associated with fractional Fourier transform, *Signal Process*, 89, 2009.
- [23] G.L. Xu, X.G. Xu, X.T. Wang, Generalized uncertainty relations of Tsallis entropy on FrFT, *Signal Image Video Process.* 15, 2021.
- [24] X. Chen, P. Dang, W.X. Mai, L^p -type Heisenberg-Pauli-Weyl uncertainty principles for fractional Fourier transform, *Appl. Math. Comput.* 491 (2025) 129236.
- [25] Q. Feng, B.Z. Li, J.M. Rassias, Weighted Heisenberg-Pauli-Weyl uncertainty principles for the linear canonical transform, *Signal Process*, 165, 2019.
- [26] P. Dang, G.T. Deng, T. Qian, A tighter uncertainty principle for linear canonical transform in terms of phase derivative, *IEEE Trans. Signal Process* 61 (21) (2013) 5153–5164.
- [27] W.X. Mai, P. Dang, W.L. Pan, X. Chen, Uncertainty principles for linear canonical transform, *J. Math. Anal. Appl.* 548 (2) (2025) 129414.
- [28] H.Y. Huo, Uncertainty principles for the offset linear canonical transform, *Circuit. Syst. Signal Process* 38 (1) (2019) 395–406.
- [29] A. Elgargati, R. Daher, Uncertainty principles for the offset linear canonical transform, *J. Pseudo-Differ. Oper. Appl.* 12 (3) (2021) 47.
- [30] W.B. Gao, B.Z. Li, Uncertainty principles for the windowed offset linear canonical transform, *Int. J. Wavelet. Multiresolut. Inf. Process* 20 (01) (2022) 2150042.
- [31] X.Y. Zhi, D.Y. Wei, W. Zhang, A generalized convolution theorem for the special affine Fourier transform and its application to filtering, *Optik (Stuttg.)* 127 (5) (2016) 2613–2616.
- [32] A. Stern, Sampling of compact signals in offset linear canonical transform domains, *Signal Image Video Process*, 2007.
- [33] J.Y. Chen, B.Z. Li, Fast numerical calculation of the offset linear canonical transform, *J. Opt. Soc. Am A* 40 (3) (2023) 427–442.